Lesson 1: why study tree rings?

This is an introductory course for learners new to dendrochronology. No previous training in tree-ring measurement or statistical analysis is required to begin. Slides 1–8 and chapter 1 explore the questions that dates can help answer. Use the video at your own pace for preparation or revision. Detailed mathematics and statistics are optional supplementary reading.

Learning goals

- Explain why tree rings are useful for investigating cultural objects, climate and forests.
- Distinguish the order of events from an explanation of their cause.
- Distinguish ring formation, felling, construction and reuse.

Why the research matters

Placing wood on a timeline helps us compare it with evidence of construction and repair. Investigating growth records helps us examine environments before instrumental observations and changes in forests. Separate what a ring dates from the event being investigated, and combine the date with other evidence.

Check your understanding

1. If you know when a ring formed, do you also know when a building containing that timber was constructed? What else would you check?

2. In a fictional example, repairs date to 1212–1215 and forest change to 1232–1235. Which came first, and how far apart might they be? Can you identify the cause?

Reasoning and review

Question 1

Ring formation need not coincide with construction. Consider subsequent growth, felling, storage and reuse. Check whether the outer edge or bark survives, where the measured ring lay within the tree, and how the timber relates to evidence about the building.

Question 2

Repairs come first in these fictional ranges. The shortest separation is $1232 - 1215 = 17$ years; the longest is $1235 - 1212 = 23$ years. Their order does not establish causation. Evidence about harvesting or land use would help test an explanation. These ranges are illustrative, not a specified probability distribution or confidence level.

Review

Describe one question tree-ring research could help answer and one conclusion that a date alone would not justify.

Introductory reading

- Speer, J. H. (2010). Fundamentals of Tree-Ring Research. Chapter 1, pp. 1–6.
- Overview of dendrochronology
- Crossdating
- Tree-ring research and an example
- Public tree-ring data

Chapter 2: classifying dating methods

For slides 9–14 and the chapter 2 video. Begin with the event you want to investigate.

Different perspectives

Perspective	Example	Question
Relative age	Stratigraphy, typology	Which event came first?
Numerical age	Tree rings, calibrated radiocarbon	What does the age or range represent?
Direct or indirect relationship	Target object, related deposit	How does the sample constrain the event?
Record or clock	Growth, decay, stored signals, magnetism	What starts and preserves the record?

The first two rows concern how age is expressed. The third concerns evidence connecting a sample to an event. The fourth concerns the mechanism. One measurement can be described from all these perspectives.

Terminology

You will encounter the term absolute dating. It does not promise an age without uncertainty. Always identify the time scale, assumptions and event represented by a result.

Clocks and events

Ring formation, mineral cooling, sand burial and the heating of a kiln are different events. Establish how the sampled record relates to the history of the site or object.

Chapter 2: check your understanding

1. Compare: this layer is older than the layer above it, and this layer is approximately a thousand years old. What information does each provide? What conditions justify the stratigraphic inference?

2. In an idealized decay model, one quarter of the original amount remains. How many half-lives have elapsed? What else would you need to convert the result into calendar time?

3. A carbonate layer formed over a deposit containing a stone tool. Does dating that carbonate directly establish the tool's manufacture? Why record the sample's position?

4. An excavation contains a charred seed and wood from deep inside a large timber. Must carbon incorporation coincide with the activity you want to date?

Chapter 2: reasoning

1. The first gives relative order. The second expresses a numerical age. Check for overturning, mixing and later disturbance. Order alone does not determine calendar years or the interval between events.

2. Two half-lives: half the original amount, followed by half the remainder. Conversion to years requires the half-life and an assessment of starting conditions, material exchange and the event that started the clock. This exercise uses an ideal model.

3. The measurement concerns carbonate formation. If the tool and strata retain their original relationship, its presence predates the overlying carbonate. That does not determine an exact year of manufacture.

4. The seed may have a short interval between growth and the activity. An inner ring may substantially predate felling or use. Investigate context, storage and reuse. Radiocarbon calibration and interpretation of the archaeological event are separate steps.

Reading

- Chronological dating
- Relative dating
- Absolute dating and uncertainty
- Radiocarbon samples and interpretation
- Radiocarbon calibration
- Scientific dating methods and sample context
- Archaeomagnetism

Dating methods and exercises

For slides 15–27 and related review.

Learning objectives

- Choose a method for the sample and event, and explain what it dates.
- Distinguish age uncertainty from temporal resolution.
- Separate dating an environmental archive from interpreting its proxy information.

Methods and targets

Method	Sample or clock	Interpretation
Dendrochronology	Ring-pattern matching	Ring formation versus felling and use
Radiocarbon	Radiocarbon in biological material	Calibration, contamination and carbon source
K–Ar / Ar–Ar, U–Th	Isotopes in minerals or carbonates	Clock closure, retention and initial components
Luminescence	Signals reset by light or heat	Resetting and burial environment
Archaeomagnetism	Magnetism related to firing and cooling	Regional reference and later movement

Age notation

1000 ± 30 BP illustrates a conventional radiocarbon age, referenced to 1950. Calibration relates it to calendar ages such as cal BP or cal BCE/CE. Subtracting an uncalibrated BP value from 1950 does not establish a calendar year. Crossdated rings receive verified calendar years.

Numerical examples and schematic diagrams are teaching examples, not measurements from actual sites.

Exercise Samples and dating methods

Discuss the cases and record your own answers. Identify what the method dates and the additional information needed.

These are fictional samples.

A Charred seeds from a hearth

Date the activity involving fire.

Possible method / event dated / additional checks

B A timber with its outer wood removed

Date construction or repair.

Possible method / event dated / additional checks

C Fired material in an apparently undisturbed kiln

Date the last firing.

Possible method / event dated / additional checks

D Buried wind-blown sand

Date deposition of the sand.

Possible method / event dated / additional checks

Key terms for Lesson 1

Review the terms below. Consult the English glossary for terms used in later lessons.

English	Brief explanation
dating	Placing a sample or event in time.
chronology	The temporal ordering of evidence and events.
relative age	An older-or-younger relationship to other evidence.
numerical age	An age expressed numerically, for example in years.
calendar age	An age placed on a calendar timescale.
temporal resolution	The time interval resolved by a record.
uncertainty	The limited certainty of a measurement or inference.
radiocarbon dating	Dating based on radiocarbon in a sample.
calibration	Relating radiocarbon ages to calendar ages using a calibration curve.
half-life	The time for half the atoms of a radioactive isotope to decay.
stratigraphy	The study of strata and their relationships.
luminescence dating	Dating time since mineral signals were reset by light or heat.
palaeomagnetism	Records of past geomagnetic fields preserved in materials.
dendrochronology	The study of change using calendar-dated tree rings.
proxy	Evidence used to infer an environmental variable indirectly.
paleoclimatology	The study of past climates.
varve	A sedimentary layer verified to represent one year.

Resources and further study

Click the linked titles. Begin with an introduction, then methods and primary sources, and finally open data. Links checked on 22 October 2026.

R01 Introduction to dating Smithsonian Human Origins Program

Introductory explanation Relative dating, stratigraphy and dating methods

R02 Geologic and radiometric time U.S. Geological Survey

Introductory explanation Radioactive decay and dating rocks

R03 Radiocarbon methods and results University of Oxford ORAU

Methods and primary explanation Samples, measurement and calibration

R04 Radiocarbon calibration University of Oxford ORAU

Methods and primary explanation BP and calendar ages

R05 Scientific dating methods Historic England

Methods and primary explanation U–Th, luminescence and palaeomagnetism

R06 Dendrochronology and crossdating U.S. National Park Service

Introductory explanation Tree-ring matching and archaeology

R07 What are proxy data NOAA NCEI

Introductory explanation Tree rings, ice, corals and pollen

R08 International Tree-Ring Data Bank NOAA NCEI

Open data Ring-width, density and isotope records

R09 Glossary of Dendrochronology WSL

Terminology and methods Definitions of technical terms in English

R11 Paleoclimate data search NOAA NCEI

Open data Search multiple kinds of natural archive

Review and preparation

Choose one exercise case and explain the target event and the limits of its date. Read the biology, mathematics and earth-science sections of BG01 before observing cambium, earlywood and latewood in Lesson 2.